

The Future of Shopping: How AI Agents Will Transform Commerce

A trustworthy, decentralized marketplace where AI assistants can actually buy things for you

[!NOTE] About This Document

This document series is an independent architectural exploration and reference implementation built upon Google's UCP specification and complementary open protocols (MCP, A2A, W3C DIDs/VCs). This is not an official Google publication. We welcome community feedback and collaboration.

What this document is NOT:

- An official Google publication or endorsed implementation
- A production-ready specification
- A commercial product

There's a scene that plays out millions of times a day. You need something — a new laptop, the latest smartphone, a birthday gift for your niece. You open a browser, hop between tabs, compare prices, read reviews, second-guess yourself, and eventually (maybe) click "Buy." The whole process takes anywhere from twenty minutes to several days. It's exhausting.

Now imagine a different world. You tell your AI assistant: *"I need a new laptop, something powerful for development but under \$1500."* Ten minutes later, your assistant confirms the order. The laptop is on its way. You didn't compare prices because your AI did. You didn't read spec sheets because your AI analyzed hundreds of them. You didn't worry about fake products because your AI verified the merchant's identity before placing the order.

This is Agentic Commerce.

The Problem with Today's E-Commerce

Today's online shopping experience was designed for humans clicking through web pages. But as we offload more decisions to AI assistants, a fundamental problem emerges: **how do you let an AI agent buy things on your behalf without trusting a central platform?**

Current solutions are deeply flawed:

The Platform Trap. Amazon, Alibaba, and other marketplaces aggregate trust centrally. Your AI could theoretically use their APIs, but you're locked into their ecosystem. They see everything, control everything, and take a cut of everything.

The Bot Problem. Most websites actively block automated access. They're designed for human browsers, not AI agents. Even if you could technically automate purchases, you'd be violating terms of service.

The Trust Gap. When AI agents transact with each other, who verifies identity? How do you know a "merchant agent" actually represents a legitimate business? How do you prove you authorized a purchase?

The question becomes: *What if we built e-commerce from scratch, specifically for the AI agent era?*

Introducing Agentic Commerce

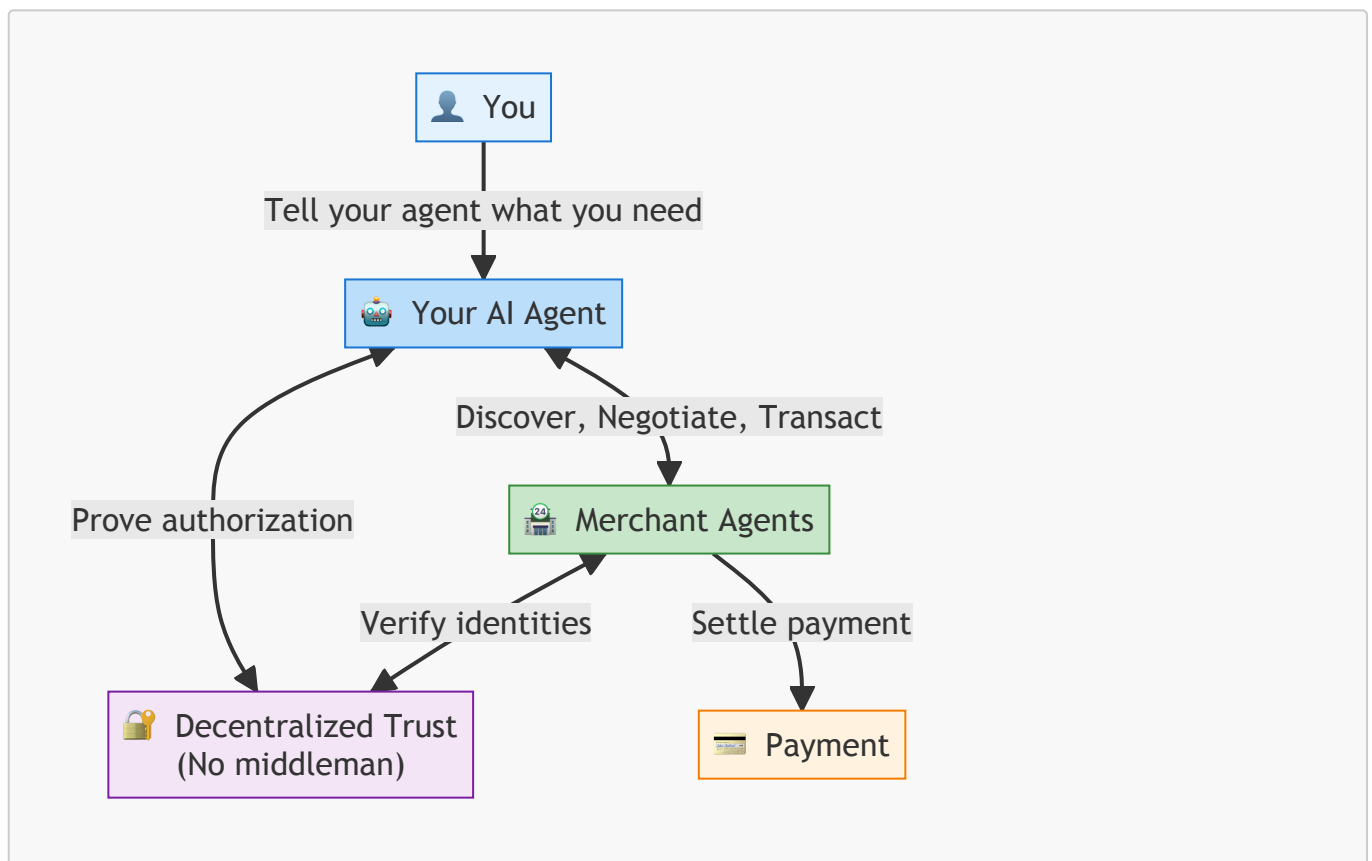
Agentic Commerce is a new paradigm where AI agents conduct transactions on behalf of users — not through hacks or workarounds, but through a purpose-built stack of protocols. The goal isn't to automate today's web shopping. It's to replace it with something fundamentally better.

At the heart of this architecture are three protocols working together:

1. **MCP** (Model Context Protocol) — How agents discover and use services
2. **UCP** (Unified Commerce Protocol) — The commerce "handshake" between buyer and seller
3. **A2A** (Agent-to-Agent Protocol) — Direct collaboration between agents

Think of it like this: MCP is the phone system that lets agents call each other. UCP is the formal contract they sign during a transaction. A2A is the backchannel where they coordinate complex tasks.

Here's the high-level picture:



Let's walk through each protocol.

MCP: The Universal Connector

The Model Context Protocol solves a basic problem: how does an AI agent know what a service can do?

Today, if you want ChatGPT to search the web, someone has to build a specific "web search plugin." If you want it to query a database, someone builds a "database plugin." Every integration is custom. Nothing is

standardized.

MCP changes this. It's a universal protocol that lets any AI agent discover and use any compatible service. A merchant running an MCP server exposes their capabilities — product search, inventory check, order placement — in a standardized format. Any AI agent speaking MCP can automatically discover and use these capabilities.

Here's what a typical MCP interaction looks like:

```
Agent: "What tools do you have?"
Merchant Server: "I have product_search, check_inventory, and
create_order."

Agent: "Search for laptops under $1500 with 32GB RAM."
Merchant Server: [Returns matching products with prices and details]

Agent: "Create an order for product LAPTOP-PRO-16."
Merchant Server: [Processes order and returns confirmation]
```

The beauty is that the agent doesn't need custom code for each merchant. If a new store launches with an MCP server, existing agents can immediately interact with it. No integration work required.

Consider an ecosystem with multiple merchant servers — one for sportswear, another for electronics — each exposing different products through identical MCP interfaces. A consumer's agent can seamlessly query both, compare prices, and choose the best option.

UCP: The Commerce Handshake

MCP handles communication, but commerce requires something more: **trust**. When your AI agent wants to buy a laptop from a merchant's AI agent, how do both sides know they're dealing with legitimate parties? How do you prove you authorized a specific purchase at a specific price?

This is where the Unified Commerce Protocol comes in. UCP defines a three-step "handshake" for every transaction:

Step 1: Intent

The buyer agent creates an **Intent Mandate** — a cryptographically signed document saying "I want to buy a development laptop, my budget is \$1500." This mandate includes the buyer's Decentralized Identifier (DID), a globally unique ID that verifies their identity.

Think of it like walking into a store and telling the clerk what you're looking for, while also showing your ID.

Step 2: Cart

The merchant agent receives the intent, checks inventory, and responds with a **Cart Mandate** — a cryptographically signed offer. "I have the ProBook 16 Developer Edition for \$1,299. This price is locked for

30 minutes." The cart mandate includes the merchant's DID, proving the offer comes from a legitimate business.

This is like the clerk handing you a formal quote on company letterhead, with their signature guaranteeing the price.

Step 3: Payment

The buyer agent reviews the cart (or more likely, had it reviewed by the user) and creates a **Payment Mandate** — authorization to complete the purchase at the agreed price. This mandate references the specific cart, ensuring the merchant can't swap in different terms.

Once the merchant receives the payment mandate, they process the order and return a confirmation — also cryptographically signed.

The key insight is that every step creates a **verifiable, non-repudiable record**. The buyer can't claim they never authorized a purchase — their signature proves they did. The merchant can't claim they offered a different price — their signature locks the terms. If there's ever a dispute, both parties have cryptographic proof of exactly what was agreed.

Decentralized Trust: No Middleman Required

Traditional e-commerce requires a trusted middleman. Amazon, eBay, and PayPal all play this role — they verify identities, hold payments, and mediate disputes. Without them, how would you know if a random website is legitimate?

Agentic Commerce takes a different approach: **decentralized identity**.

Every participant — every user, every merchant, every agent — has a Decentralized Identifier (DID). Unlike usernames or email addresses, DIDs are:

- **Self-sovereign**: You control your identifier; no platform can revoke it
- **Cryptographically verifiable**: Anyone can verify that a DID belongs to who it claims to
- **Globally unique**: No two entities share the same DID

DID Methods: Flexibility in Implementation

The beauty of DIDs is that they're method-agnostic. Different use cases call for different approaches:

- **did:web** — Uses your domain as identity anchor. A merchant at electronics.com gets did:web:electronics.com. Simple to set up, leverages existing web infrastructure, and naturally maps to business identity. If you own the domain, you control the DID.
- **did:key** — Self-contained cryptographic identity. The DID itself encodes the public key, requiring no external resolution. Perfect for ephemeral agents, quick proofs of concept, or scenarios where you want zero infrastructure dependencies.
- **did:ethr** — Anchored to Ethereum blockchain. The DID maps to an Ethereum address, with key management on-chain. Ideal when you want immutable, censorship-resistant identity with full decentralization. No company can revoke your identity because it lives on a public blockchain.

A consumer might use `did:key` for simplicity. A major retailer might prefer `did:web` tied to their corporate domain. A decentralization purist might insist on `did:ethr` for maximum trustlessness. The protocol works with all of them.

When merchants sign their Cart Mandates, they use the private key associated with their DID. Anyone can verify the signature without calling a central authority. It's like a digital notary that works everywhere, instantly, for free.

All mandates are packaged as Verifiable Credentials (VCs) — a W3C standard for cryptographically signed claims. A Cart Mandate VC states: "Merchant DID:xyz offers Product ABC for \$1,299." Anyone can verify this claim by checking the signature against the merchant's DID.

The result is a trust system that works without intermediaries. Merchants prove their identity by signing offers. Buyers prove authorization by signing mandates. Neither needs to trust a middleman, because cryptography provides the guarantees.

A Real Transaction: Step by Step

Let's walk through an actual purchase:

The user says: "Find me a laptop under \$1500, 32GB RAM, good for coding."

1. The consumer agent creates an Intent Mandate VC, signed with the user's DID
2. It sends this intent to a merchant's MCP server via the `ucp_intent` tool
3. The merchant's server verifies the user's signature, confirming the request is authentic
4. The merchant checks inventory and finds "ProBook 16 Developer Edition" for \$1,299
5. The merchant creates a Cart Mandate VC, signed with their DID
6. The consumer agent receives the cart and verifies the merchant's signature

The user says: "Buy it."

7. The consumer agent creates a Payment Mandate VC, referencing the cart and the total amount
8. It sends this to the merchant via the `submit_payment_mandate` tool
9. The merchant verifies the signature, confirms the cart is still valid
10. The merchant processes the payment and creates the order
11. The merchant sends an Order Confirmation VC back to the consumer agent
12. The agent confirms to the user: "Order placed! #ORD-12345"

At every step, signatures are verified. At every step, both parties have cryptographic proof of what happened. And crucially, no central platform was involved.

A2A: When Agents Need to Collaborate

Commerce often involves more than two parties. What if a purchase requires coordination between shipping companies, payment processors, and multiple merchants? This is where Agent-to-Agent (A2A) protocol comes in.

A2A enables direct communication and task delegation between agents. Each agent publishes an "Agent Card" describing its capabilities — essentially a business card for AI. When agents need to collaborate, they

can:

- Discover each other's capabilities
- Delegate subtasks
- Share progress updates
- Coordinate complex workflows

Imagine a consumer agent that needs to find the best deal across multiple merchants. Rather than the user manually comparing, the agent queries merchant agents simultaneously, receives offers, and makes a recommendation. Shipping agents, tax agents, and logistics agents can all coordinate behind the scenes to fulfill complex orders.

Why This Matters

Agentic Commerce isn't just a technical curiosity. It's a glimpse of how the AI economy might actually work.

For consumers: Shopping becomes effortless. Instead of spending hours comparing products, you delegate to an agent that handles everything — and you get cryptographic guarantees that it did what you asked.

For merchants: No more platform fees. No more fighting for algorithmic visibility. If you run an MCP server, any agent in the world can find you and buy from you. The playing field levels.

For the ecosystem: Decentralization means resilience. No single company controls the marketplace. No platform can arbitrarily change the rules. Commerce becomes as open as the web itself.

Conclusion: Commerce for the AI Era

The transition to AI agents won't happen by making agents pretend to be humans clicking through websites. It will happen by building new infrastructure — protocols, standards, and systems — designed from the ground up for machine-to-machine commerce.

Agentic Commerce represents that infrastructure. Three protocols working together: MCP for discovery, UCP for transactions, and A2A for collaboration. Built on decentralized identity, secured by cryptography, open to anyone.

The future of shopping isn't clicking "Buy Now." It's telling your agent what you want and trusting that it will handle the rest — with proof.

Acknowledgments

This work builds upon open protocols and standards developed by leading organizations:

- **UCP** (Unified Commerce Protocol) — Google
- **MCP** (Model Context Protocol) — Anthropic
- **A2A** (Agent-to-Agent Protocol) — Google
- **DIDs & VCs** — W3C

Feedback Welcome: Community feedback and contributions are encouraged.